

Position History

- 09/25 - present Scholarly Engagement Lead, Strategic Engagement Leads Team, Scientific Data Division, Lawrence Berkeley National Laboratory, Berkeley, CA
- 12/23 – 09/25 Interim Group Leader, Scientific Data Management Group, Scientific Data Division, Lawrence Berkeley National Laboratory, Berkeley, CA
- 07/13 - present Senior Computing Engineer, Scientific Data Division
Lawrence Berkeley National Laboratory (LBNL), Berkeley, CA
- Recent research area includes in-network data caching strategies, performance diagnostics and modeling, dynamic resource management, and statistical modeling and machine learning methods for autonomous scientific data infrastructure.
 - Contributed to over 350 paper publications and technical reports, a number of open source software packages and standard specifications, and multiple patented technologies, in data, cloud computing, HPC, and networking areas.
- 10/02 - 06/13 Computer Science Engineer IV, Computational Research Division, LBNL, Berkeley, CA
- 04/00 - 10/02 Computer Science Engineer III, Computational Research Division, LBNL, Berkeley, CA
- 03/00 - 07/00 Technical Adviser, NetGeo, Inc, Mountain View, CA
- 07/97 - 04/00 Computer Science Engineer II, National Energy Research Scientific Computing Center, Lawrence Berkeley National Laboratory, Berkeley, CA
- 10/96 - 11/98 Co-founder, IVision Corporation, Alameda, CA
- 08/95 - 06/97 Artificial Intelligence Software Engineer, Stottler Henke Associates Inc., San Mateo, CA
- 03/94 - 08/95 Graduate Research Assistant, Imaging Technologies Group, LBNL, Berkeley, CA
- 08/90 - 05/93 Graduate Student Instructor, University of California at Berkeley, Berkeley, CA

Selected Publications

- V. Yakubovich, I. Sim, P. Kim, N. Monozon, J. Chung, A. Sim, K. Wu, "Time-series Forecasting for Network Utilization in Large-Scale Scientific Workflows", IEEE International Conference on Computing, Networking and Communications (ICNC 2026), 2026. Best Paper Award.
- A. Sim, E. Wang, R. Monga, K. Wu, C. Guok, I. Monga, D. Davila, F. Wurthwein, J. Balcas, H. Newman, "Comparing Cache Utilization Trends for Regional Data Caches", European Physical Journal (EPJ) Web of Conferences (CHEP 2024), Vol. 337, No. 01341, 2025.
- R. Shao, A. Sim, K. Wu, J. Kim, "Leveraging History to Predict Abnormal Transfers in Distributed Workflows", Sensors, 23(12), pp. 5485, MDPI, 2023.
- B. Weinger, J. Kim, A. Sim, M. Nakashima, N. Moustafa, K. Wu, "Enhancing IoT anomaly detection performance for federated learning", Digital Communications and Networks, 8(3):314-323, 2022.
- M. Nakashima, A. Sim, Y. Kim, J. Kim, J. Kim, "Automated Variable Selection for Network Anomaly Detection", ACM Transactions on Management Information Systems (TMIS), 2021.
- J. Kim, A. Sim, B. Tierney, S. Suh, I. Kim, "Multivariate Network Traffic Analysis using Clustered Patterns", Journal of Computing, 101(4):339-361, 2019.
- J. Wang, W. Yoo, A. Sim, P. Nugent, K. Wu, "Parallel Variable Selection for Effective Performance Prediction", IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid), 2017.
- D. Lee, A. Sim, J. Choi, K. Wu, "Novel Data Reduction Based on Statistical Similarity", 28th International Conference on Scientific and Statistical Database Management (SSDBM), 2016.
- W. Yoo, A. Sim, "Time-series Forecast Modeling on High-Bandwidth Wide Area Network Measurements", Journal of Grid Computing, 14(3):463-476, 2016.
- K. Hu, J. Choi, A. Sim, J. Jiang, "Best Predictive Generalized Linear Mixed Model with Predictive Lasso for High-Speed Network Data Analysis", International Journal of Statistics and Probability, 4(2):132-148, 2015.

Selected Grants

1. *AMASE: Architecture and Management for Autonomic Science Ecosystems*, DOE ASCR, Smart High-Performance Computing and Communications Systems Program, 9/2017-8/2021, Co-PI, \$4.2M
2. *IDEALEM, Novel Data Reduction Based on Statistical Similarity*, LBNL LDRD, CS Innovation Project, 10/2015-9/2016, PI, \$100K
3. *Scientific Data Services (SDS) – Autonomous Data Management on Exascale Infrastructure*, DOE ASCR, 7/2015-6/2018, Co-PI, \$2.4M
4. *Scalable Analysis Methods and In Situ Infrastructure for Extreme Scale Knowledge Discovery*, DOE ASCR, 10/2014-9/2017, Co-PI, \$4.5M
5. *Policy-driven Large Scale Data Access Framework*, NSF Collaborative Research SDCI Net, 9/2011-8/2014, Co-PI, \$1.2M

Selected Software Package Releases

1. *Network Resource Manager for OSCARS*, 2018-2021.
2. *Implementation of Dynamic Extensible Adaptive Locally Exchangeable Measures (IDEALEM)*, 2016-2025.

Selected Activities

Technical Program/Standard Committee: AI-Science, AICT, BDCAT, CCGrid, CCSNA, Cloud Computing, CloudNet, Cluster, DADC, Data Analytics, DDINS, DIDC, FGCN, GDC, Globecom, Grid, HPC Asia, ICC, ICCN, ICNS, ICONS, ISAV, ISCC, NAS, NDM, NMIC, OASIS WSRF, OASIS WSN, OGF GSM, SC, SioTEC, SNTA, UCC, VIRT

Chair: AI-Science (ACM), HPC Asia, SNTA (ACM), SNTA (IEEE)

Steering Committee: Cloud Computing, ICNS, SNTA, WATCH

Journal editorial board: Frontiers in HPC, IJBDI, IJGDC, Sensors

Founding member: IEEE SIG-IIE, OGF GSM-WG

Selected Awards, Honors and Patents

1. Director's Award for Exceptional Scientific Achievement, LBNL, 2023.
2. US Patent 10,908,972 B2, "*Methods, systems, and devices for accurate signal timing of power component events*", 2021.
3. US Patent 10,366,078 B2, "*Data reduction methods, systems, and devices*", 2019.
4. US Patent 8,705,342 B2, "*Co-scheduling of network resource provisioning and host-to-host bandwidth reservation on high-performance network and storage systems*", 2014.
5. Best paper award (ICNC 2026, DataCom 2015, ISC 2005)
6. Federal Laboratory Consortium For Technology Transfer, FAR-WEST Region, Outstanding Partnership Award, *Earth System Grid Federation*, 10/2013.
7. ACM Student Research Competition (SRC): 1st place SC'23, 2nd place SC'22, 1st place TAPIA'19, 1st place SC'17, 2nd place SC'16, 3rd place SC'16, 3rd place SC'10.
8. Hottest Infrastructure Award, A Data Management Infrastructure for Climate Modeling Research – Super Computing 2000 Network Challenge (2000).
9. Outstanding Performance Award – Lawrence Berkeley National Laboratory (1999, 2001).
10. High Distinction in General Scholarship, Departmental Honors in Statistics – U.C. Berkeley (1990).

Education

- M.S. in Computer Science, San Francisco State University, August 1995
- M.A. in Statistics, University of California at Berkeley, May 1993
- B.A. (High Distinction in General Scholarship, equiv. to Magna Cum Laude) in Applied Mathematics and Statistics (Double Major), University of California at Berkeley, May 1990

Miscellaneous:

U.S. Citizen
 U.S. DoD security clearance – secret level (1995-1997)
 IEEE Senior Member, ACM Member, and Member of Society of Actuaries
 Full CV is available upon request